

How different are two probabilistic programs?

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Contextual Distance

Contextual Equivalence (Morris 1969)

$$M \sim^{ctx} N \quad \text{iff} \quad \forall C, \text{Obs}(C[M]) = \text{Obs}(C[N]).$$

- **Contexts** : Model the environments.
- **Observation** :
 - **Deterministic case** : the execution result is 1 ?
 - **Probabilistic case** : probability of termination with 1.

$$M \oplus N \begin{array}{l} \xrightarrow{1/2} M \\ \xrightarrow{1/2} N \end{array}$$

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Two programs behave in a **similar** way

$$\left. \begin{array}{l} \bullet M = \underline{1} \\ \bullet N = \underline{0} \oplus^{\epsilon} \underline{1} \text{ with } \epsilon \ll 1 \end{array} \right\} \begin{array}{l} C[\cdot] = [\cdot] \\ \text{Proba}([M] \downarrow 1) = 1 \\ \text{Proba}([N] \downarrow 1) = 1 - \epsilon \end{array} \quad M \not\sim^{ctx} N$$

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Question : How to generalize contextual equivalence in order to compare quantitatively two programs ?

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Question : How to generalize contextual equivalence in order to compare quantitatively two programs ?

Contextual Distance

$$d^{ctx}(M, N) = \sup_{C \text{ context}} |\text{Proba}(C[M] \downarrow 1) - \text{Proba}(C[N] \downarrow 1)| \in [0, 1].$$

Trivialization Phenomenon : An Example

Very close programs :

- $M = \underline{1}$,
- $N_\epsilon = \underline{0} \oplus^\epsilon \underline{1}$, where $\epsilon \ll 1$.



Problem

$$d^{ctx}(M, N_\epsilon) = 1!$$

Computation of $d^{ctx}(M, N)$:

Construction of $C_2[\cdot]$:

- **copies** its argument 2 times
- **evaluates** all copies
- **returns** 1 if all instances return 1, 0 otherwise.

Example :

- $\text{Proba}(C_2[M] \downarrow 1) = 1$.
- $\text{Proba}(C_2[N_\epsilon] \downarrow 1) = (1 - \epsilon)^2$.

$$\begin{aligned} \forall \epsilon > 0, \quad d^{ctx}(M, N_\epsilon) &\geq |\text{Proba}(C_2[M] \downarrow 1) - \text{Proba}(C_2[N_\epsilon] \downarrow 1)| \\ &\geq |1 - (1 - \epsilon)^2|. \end{aligned}$$

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Problem

$$d^{ctx}(M, N_\epsilon) = 1!$$

Computation of $d^{ctx}(M, N)$:

Construction of $C_n[\cdot]$:

- **copies** its argument n times
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- $\text{Proba}(C_n[M] \downarrow 1) = 1$.
- $\text{Proba}(C_n[N_\epsilon] \downarrow 1) = (1 - \epsilon)^n$.

$$\begin{aligned} \forall \epsilon > 0, \quad d^{ctx}(M, N_\epsilon) &\geq \lim_{n \rightarrow \infty} |\text{Proba}(C_n[M] \downarrow 1) - \text{Proba}(C_n[N_\epsilon] \downarrow 1)| \\ &= \lim_{n \rightarrow \infty} |1 - (1 - \epsilon)^n| = 1. \end{aligned}$$

Amplification Contexts

In the previous example :

$$\begin{aligned} d^{ctx}(M, N_\epsilon) &\geq \lim_{n \rightarrow \infty} |\text{Proba}(C_n[M] \downarrow 1) - \text{Proba}(C_n[N_\epsilon] \downarrow 1)| \\ &= \lim_{n \rightarrow \infty} |1 - (1 - \epsilon)^n| = 1. \end{aligned}$$

Amplification family

Let M, N be two programs of the same type.

A family of contexts $(C_n)_{n \in \mathbb{N}}$ is an **amplification family for M and N** if

$$\lim_{n \rightarrow \infty} |\text{Obs}(C_n[M]) - \text{Obs}(C_n[N])| = 1.$$

Trivialization of Distance :

Theorem 1. (Crubillé, Dal Lago 2017)

Let L be a reasonable probabilistic language such that :

- *all programs terminate with probability 1,*
- *contexts can copy their arguments.*

Then, for all (M, N) , we have $d^{\text{ctx}}(M, N) \in \{0, 1\}$.

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Example : $\mathbb{T}_{\oplus}$, a probabilistic variant of Gödel's system $\mathbb{T}$.

Types :

$$\sigma, \tau ::= \text{Nat} \mid \sigma \rightarrow \tau \mid \sigma \times \tau.$$

Terms :

$$\begin{aligned} M, N ::= & x \mid \underline{n} \mid \lambda x. M \mid MN \mid M \oplus N \\ & \mid \text{succ}(M) \mid \langle M, N \rangle \mid \text{fst}(M) \mid \text{snd}(M) \\ & \mid \text{rec}(M, N, P). \end{aligned}$$

Alternative Notions of Contextual Distances

1- Restriction on the ability to copy arguments

- Linear Distance (*Crubillé, Dal Lago 2017*) :

$$d_{\text{lin}}(M, N) = \sup_{C \text{ linear}} |\text{Proba}(C[M] \downarrow 1) - \text{Proba}(C[N] \downarrow 1)|.$$

- Graded Contextual Distances :

$$\delta^n(M, N) = \sup_{\substack{C \text{ uses the} \\ \text{argument} \\ \text{at most } n \text{ times}}} |\text{Proba}(C[M] \downarrow 1) - \text{Proba}(C[N] \downarrow 1)|.$$

Alternative Notions of Contextual Distances

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Trivialization reappears in the limit :

$$\delta^1, \delta^2, \dots \longrightarrow d^{\text{ctx}}.$$

Alternative Notions of Contextual Distances

1- Restriction on the ability to copy arguments

2- Introduction of Divergence

- Amortized Distance (T. Ehrhard) :

$$\delta^r(M, N) = \sup_C \left| \text{Obs}(\textcolor{red}{C}[M \oplus^r \Omega]) - \text{Obs}(\textcolor{red}{C}[N \oplus^r \Omega]) \right|, \quad r \in (0, 1).$$

Why there is no trivialization with δ_{ctx}^r .

- Suppose there exists an A.C.F. C_n such that $\lim_{n \rightarrow \infty} |\text{Obs}(C_n([M])) - \text{Obs}(C_n([N]))| = 1$
-

$$\begin{aligned} \delta_{ctx}^r(M, N) &\geq \lim_{n \rightarrow \infty} \left| \text{Obs}(C_n([M \oplus^r \Omega])) - \text{Obs}(C_n([N \oplus^r \Omega])) \right| \\ &\geq \lim_{n \rightarrow \infty} \textcolor{red}{r}^n |\text{Obs}(C_n([M])) - \text{Obs}(C_n([N]))| \\ &= 0. \end{aligned}$$

Contributions (Mansour, Crubillé)

- An alternative proof of the trivialization phenomenon via total variation distance.
- Resource-aware contextual distances :
 - Graded distance.
- Quantitative lower bounds obtained from amplification, for both the graded and amortized distances.

How to Use Total Variation Distance ?

Recall : Contextual Distance ($\mathbb{T}^\oplus$)

$$d^{ctx}(M, N) = \sup_C |Proba(C[M] \downarrow 1) - Proba(C[N] \downarrow 1)|.$$

Total Variation Distance

Let P, Q be two probability distributions on a finite set S .

$$d_{TV}(P, Q) = \sup_{A \subseteq S} |P(A) - Q(A)|.$$

Contexts Simulating Events

Let $M : \text{Bool}$ with :

$$\text{Proba}(M \downarrow 1) = e_1 \quad \rightsquigarrow \quad \text{Bernoulli}(e_1)(1) = e_1.$$

For $n \in \mathbb{N}$ and $A \subseteq \{0, 1\}^n$,
we construct a context $C_A^n[\cdot]$.

- **copies** its argument n times
- **evaluates** all copies
- we obtains

$$(b_1, \dots, b_n) \in \{0, 1\}^n;$$

- returns 1 iff

$$(b_1, \dots, b_n) \in A.$$

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Therefore,

$$\text{Proba}(C_A^n[M] \downarrow 1) = \text{Bernoulli}(e_1)^{\otimes n}(A).$$

From Contextual Distance to Total Variation

Let $M, N : Bool$ with

$$Proba(M \downarrow 1) = e_1, \quad Proba(N \downarrow 1) = e_2.$$

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Then,

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Since this holds for every $A \subseteq \{0, 1\}^n$,

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Therefore,

$$d^{\text{ctx}}(M, N) \geq d_{TV}(\text{Bernoulli}(e_1)^{\otimes n}, \text{Bernoulli}(e_2)^{\otimes n}).$$

Amplification via Total Variation

Let $M, N : \text{Bool}$ with

$$\text{Proba}(M \downarrow 1) = e_1, \quad \text{Proba}(N \downarrow 1) = e_2.$$

$$d^{\text{ctx}}(M, N) \geq d_{TV}(\text{Bernoulli}(e_1)^{\otimes n}, \text{Bernoulli}(e_2)^{\otimes n}).$$

By probability tools, if $e_1 \neq e_2$,

$$d_{TV}(\text{Bernoulli}(e_1)^{\otimes n}, \text{Bernoulli}(e_2)^{\otimes n}) \xrightarrow{n \rightarrow \infty} 1.$$

Therefore,

$$d^{\text{ctx}}(M, N) = 1.$$

Lower Bounds

Let $M, N : \text{Bool}$ with

$$\text{Proba}(M \downarrow 1) = e_1, \quad \text{Proba}(N \downarrow 1) = e_2.$$

For a grade $p \in \mathbb{N}$,

$$\delta^p(M, N) \geq \sup_{0 \leq n \leq p} \underbrace{d_{TV}(\text{Bernoulli}(e_1)^{\otimes p}, \text{Bernoulli}(e_2)^{\otimes p})}_{\text{explicitly computable}}.$$

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Thank you for your attention !